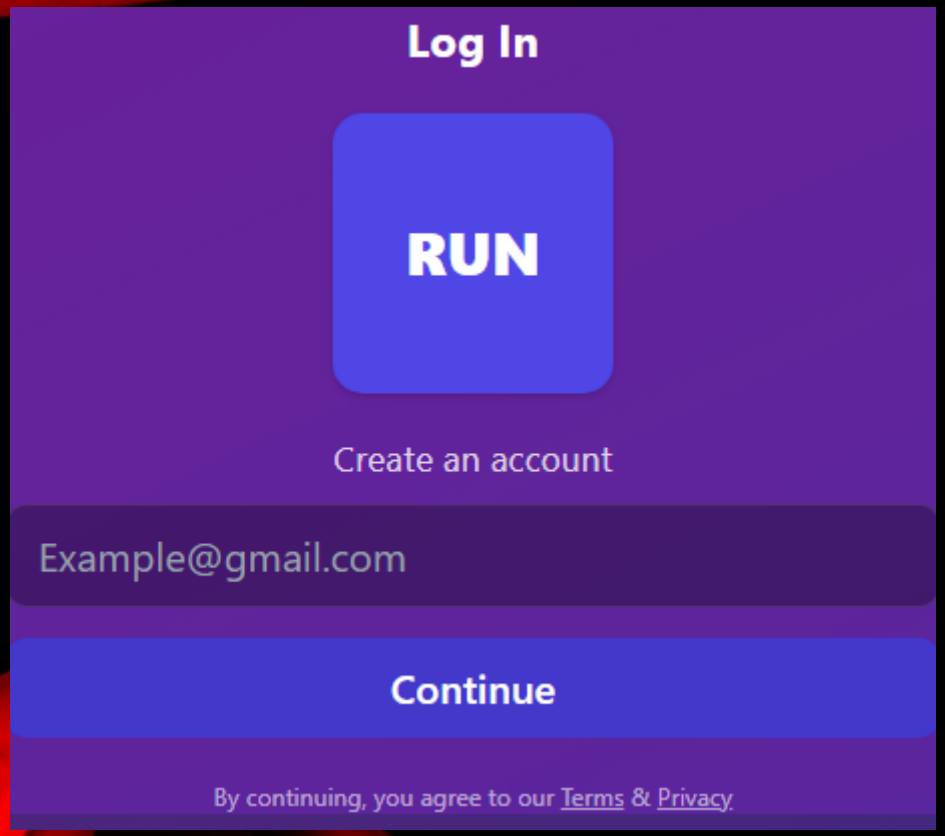
Tool Name- Workout RPG Tracker
Team Name- RUN
API Name- HealthKit / Google Fit (Primary) Alternate APIs (For Future Use): Fitbit API, Garmin API

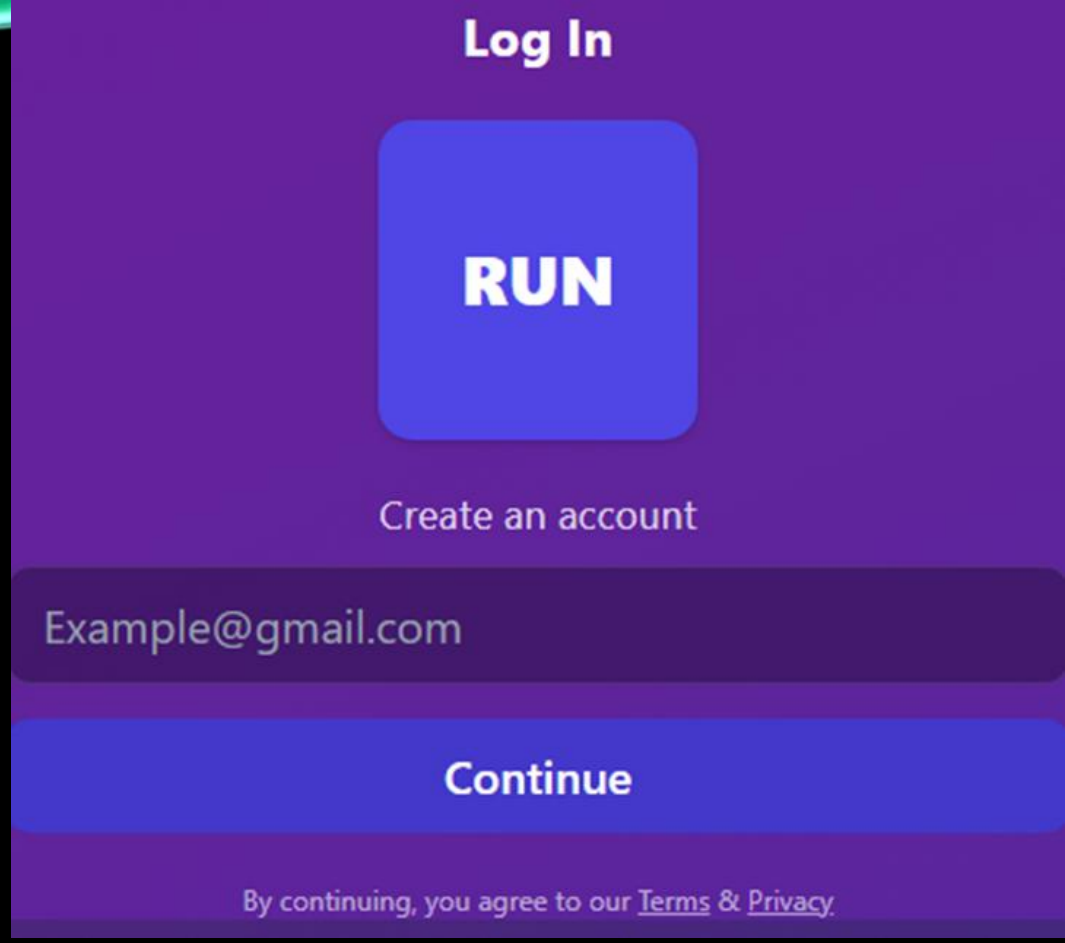


TEAM RUN
GROUP MEMBERS-

1. DANIEL HERRERA
2. CARLOS MARTINEZ
3. ANTHONY JIMENEZ
4. HARUT KULAKCHIAN
5. OSBALDO BRAVO

PROJECT SUMMARY

- Workout RPG Tracker is a fitness tracking web/mobile application that combines workout logging with role-playing game mechanics. Users log exercises such as pushups, squats, pull-ups, and hydration intake. Each activity updates character stats in real time.
- The application connects to wearable APIs (HealthKit / Google Fit) to retrieve heart rate and activity data. Logged workouts increase character attributes like strength, stamina, and agility.
- The system stores user data securely, calculates experience points (XP), and unlocks features such as boss battles, raids, customization, and progression zones based on consistency and performance.
- The goal is to increase workout motivation through a game while maintaining real fitness tracking accuracy.



SYSTEM ROLES

User (Player)

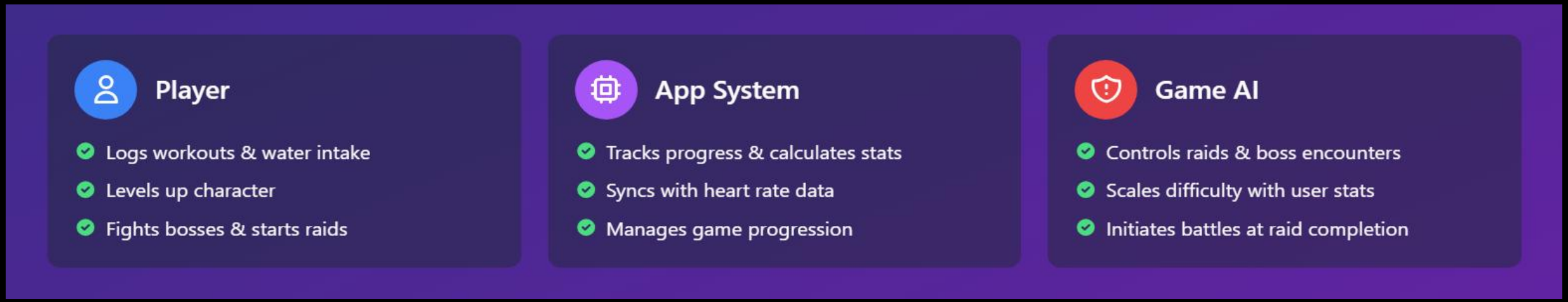
- **Creates account and logs in**
- **Logs workouts and hydration**
- **Connects smartwatch or fitness API**
- **Customizes character**
- **Participates in raids and boss battles**

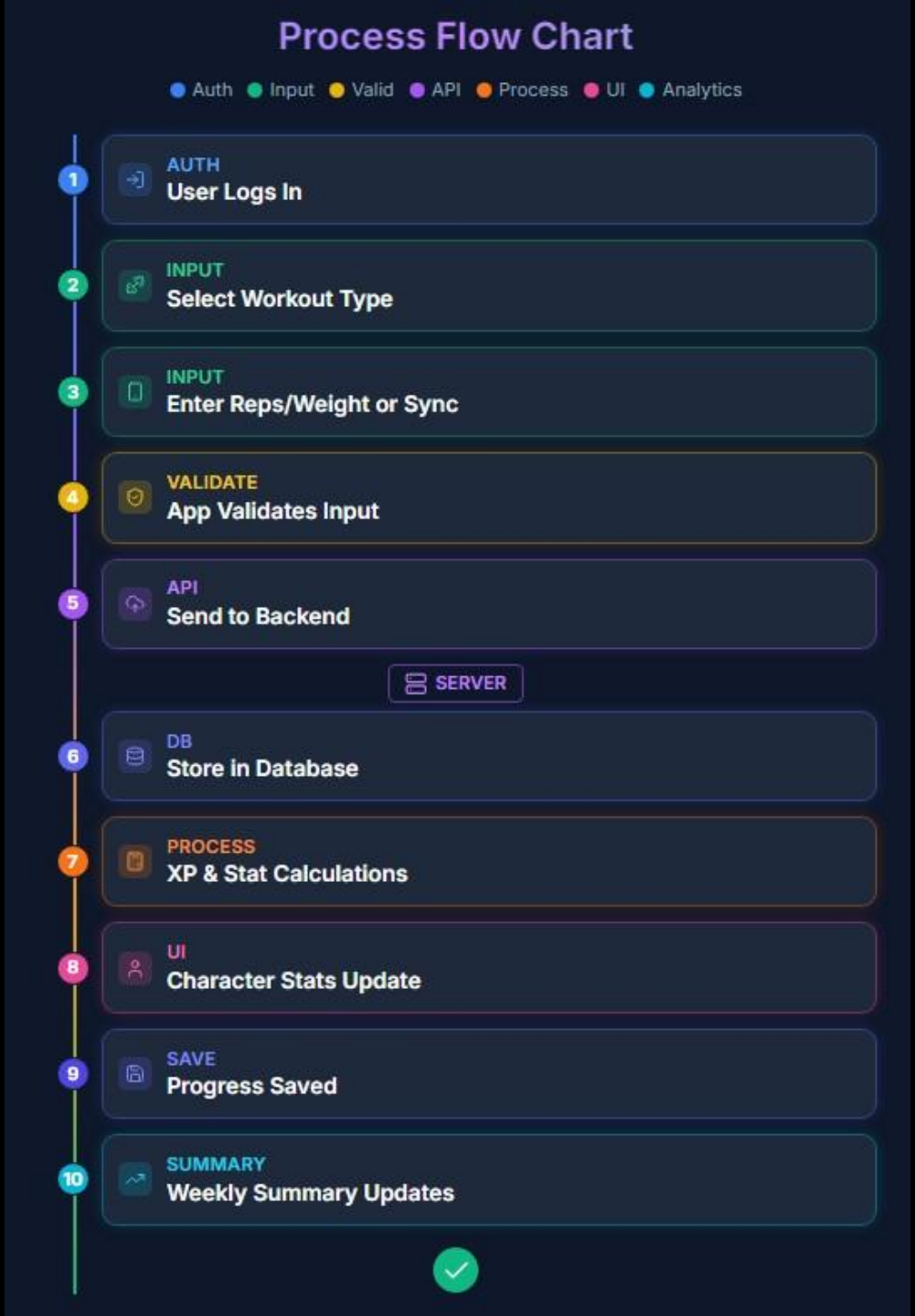
Application System

- **Authenticates users**
- **Stores and retrieves workout data from database**
- **Calculates XP and character stat increases**
- **Syncs wearable data from APIs**
- **Controls progression logic**

Game Engine / AI System

- Determines raid difficulty
- Calculates boss battle outcomes
- Applies stat multipliers
- Triggers reward system





PROCESS FLOW CHART

ACCESSIBILITY DESIGN

- Large readable fonts
- High contrast color mode
- Dark mode option
- Screen reader support
- Simple navigation layout
- Clear button labels
- Alternative text for animations
- Adjustable notification settings

Feature Demo: Character Creation

Create Your Hero

Character Name

Enter hero name

Class

Warrior | Ranger | Mage

Appearance

Starting Stats

Strength:	5
Stamina:	5
Agility:	5

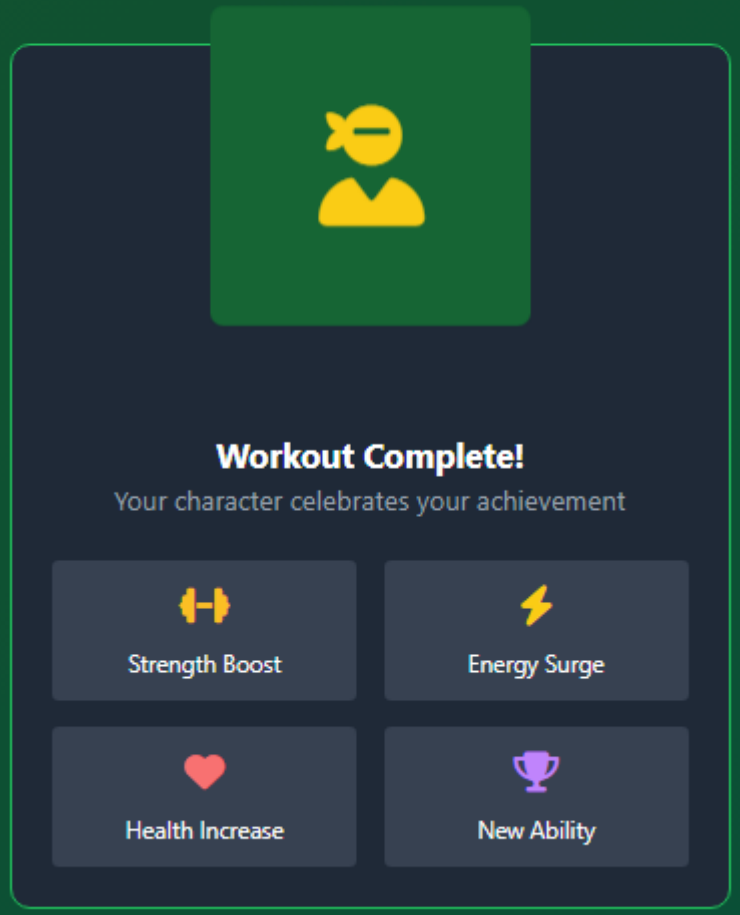
Create Hero

Your Hero

Stats will increase as you complete real-world workouts

Level: 1

Character Animations



Dynamic Exercise Animations

Your character responds to different workout types with specialized animations:

Weight Training
Character visibly strains with heavy weights

Cardio
Character runs faster as pace increases

Yoga
Character flows smoothly between poses

Animation Examples

- Pushups → Arm muscles flex
- Squats → Legs glow with power
- Cardio → Character runs in place
- New PR → Victory pose
- Level up → Full transformation

Motivational System

Animations provide visual feedback and encouragement to keep users engaged with their fitness journey.

Body Part Focus

Targeted Training

Choose which body part to focus on for bonus XP and specialized workouts.

Visual Feedback

- Selected area glows on character
- Special animations for focus area
- Visual muscle growth over time
- Progress bars for each body part

Workout Suggestions

The app recommends exercises based on your selected focus area to help you achieve balanced fitness.

Feature Demo: Workout Logging

Squats Legs

+5 Leg Strength · Energy recovery +2%

Log Your Workout

Body Part

Legs

Exercise

Squats

Reps

12

Weight (lbs)

135

Character Animation

Play Skip

Log Workout

Log Reps & Weight

Comprehensive Workout RPG System

Our advanced algorithm converts your physical efforts into character progression with:

Strength Training
Higher weights = more XP

Endurance
More reps = stamina bonuses

Cardio
Distance = movement speed

Flexibility
Yoga = agility stats

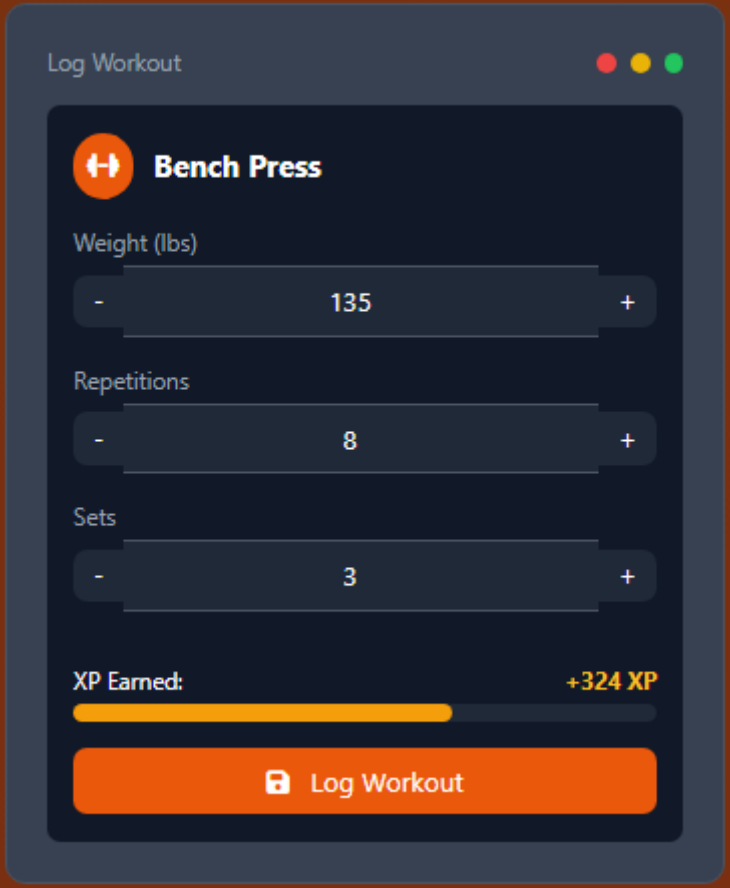
XP Calculation

```
XP = (Reps × Weight) × Exercise Multiplier
```

Different exercises have different multipliers based on intensity.

Progression Tracking

- Visual graphs of strength gains
- Milestone achievements
- Personal records tracking
- Workout history calendar



Water Intake Tracker

Hydration = Health

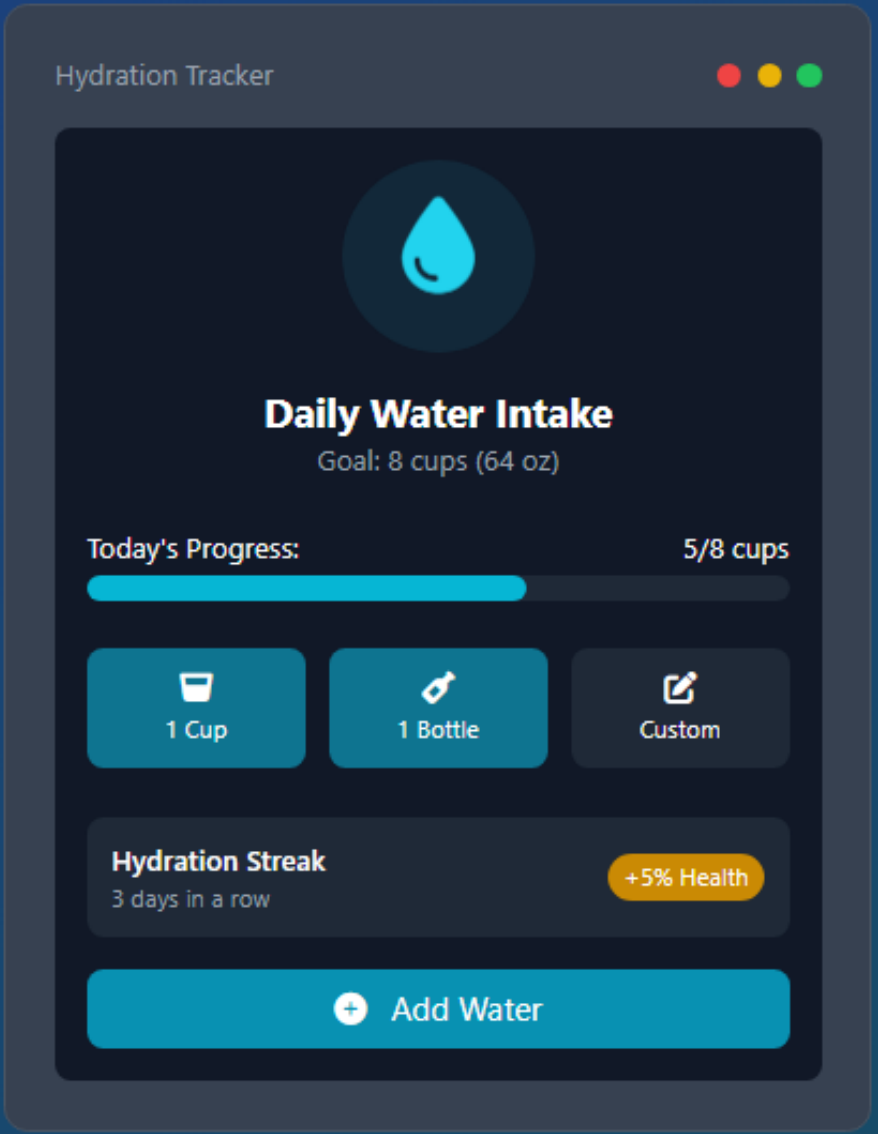
Track your water intake to boost your character's health and stamina stats.

Features

- Daily water goal tracking
- Reminder notifications
- Hydration streak rewards
- Visual character effects when hydrated
- Health stat bonuses

Health Benefits

Proper hydration improves workout recovery and overall health, reflected in your character's abilities.



Ready for Your Fitness Quest?

Workout Tracking

Turn exercises into character XP

Hydration

Water intake boosts health stats

Progression

Level up through consistency

Start Your Adventure

Transform your fitness routine into an epic RPG journey today!

USER STORIES: (EXAMPLES)

1. As someone who plays games and goes to the gym, I want a tracker that keeps me incentivized so that I stay consistently in the gym.

Acceptance Criteria:

a. User can tie progress to an account.

b. User can let built-in software have certain permissions.

c. User can create a character.

Tasks:

a. Create an account using email/password

b. Enable tracking permissions such as heart rate, steps, breathing, and water intake

c. Create and design a character in a design area UI

2. As someone who struggles with warming up, I want a plan that will have warm-ups that will keep me engaged.

Acceptance Criteria:

a. User can choose a "Warm-Up" option

b. User can choose between set warm-ups or create their own

c. User can choose to display a timer or a video for the warm-ups chosen

d. User can see a character animation do it alongside him

Tasks:

a. Make a button that will lead to a set of warm-up options such as a light warm-up, or heavy warm-up.

b. Will log the warm up and update stats accordingly

c. Displays the created character and animates the character doing the warm-ups

USER STORIES (CONTINUED):

3. As someone who forgets to track my current workout, I want to be able to log my workout in a way that keeps me consistent and honest.

Acceptance Criteria:

a. User can choose what body part

b. User can enter the number of repetitions and weight

c. User can see character stats increase as information is logged

d. User can see character do animations based on the workout

Tasks:

a. Have body parts listed that the user can choose from

b. Show workouts related to the body part that the user can choose from

c. Be able to enter a custom amount of repetitions and weights for the workout chosen

d. Play an animation of the character doing the workout chosen

e. Increase the body part stats for the created character based on the info that the user inputs

4. As someone who struggles drinking enough water, I want to be able "game-ify" the task so that I am no longer struggling with it

Acceptance Criteria:

a. User can choose a water intake option

b. User is able to choose the amount of water

c. User is able to see a water-intake progress bar

d. User can see the effect it has on character in raids/battles

Tasks:

a. Have a water icon displayed on screen that can be pressed

b. Show amounts of water (cups/bottles/ounces/etc.) that can be pressed and be able to input an amount

c. Play an animation showing the progress bar filling up

d. Have and show battle buffs from the amount of water drank

USER STORIES (CONTINUED):

5. As someone who can struggle with going to the gym, I want to see how not going can affect my character so that I have incentive to go consistently

Acceptance Criteria:

a. User can see stats lower

b. User can see chance of raid success lower

Tasks:

a. Have a custom schedule and track if user did not go and lower stats accordingly

b. Show a raid success percentage and have it lower according to how many days/what type of day is missed

6. As someone who likes seeing consistency rewarded, I want a way to visualize my own diligence so that I would want to keep going

Acceptance Criteria:

a. User can see a 3/7/15/30 day consistency pop up

b. User can see a new map zone unlocked per each goal met

c. User can visually see the map unlock through an animation

Tasks:

a. Create a pop up that displays on each goal achieved

b. Have zones blocked until each day-streak is met

c. Display an animation for each zone unlocking that plays after achieving each goal

USER STORIES (CONTINUED):

7. As someone who enjoys progressing to beat a boss, I want a way to see my progress reach a goal so that I can challenge a boss

Acceptance Criteria:

a. User can fight a boss at the end of a raid

b. User will be able to see if boss can be challenged based on stats

c. User can choose a reward after each boss beaten

d. User can choose to see or skip the battle animation and outcome

Tasks:

a. Have a boss as a battle at the end of a raid

b. The boss should be able to be challenged if the character reached the required stats

c. Cap the character reward to not be too powerful

d. Create a reward list to choose between after defeating the boss

e. Show a battle animation and outcome but let the user have the choice to skip it

8. As someone who enjoys earning rewards through effort, I want to unlock and equip gear so that I can feel my progress making my character stronger

Acceptance Criteria:

a. User can unlock gear by meeting specific goals or milestones

b. User can equip or change gear once unlocked

c. User can view achievements related to unlocking gear

d. Equipping gear should increase character stats

Tasks:

a. Track milestone achievements and display them in a side slide notification

b. Lock gear until the required stats or goals are met

c. Create a wardrobe where new items appear once unlocked

d. Implement stat boosts when gear is equipped

USER STORIES (CONTINUED):

9. As someone who wants an easier way to track workouts, I want to use my heart rate sensor so that my workouts log automatically without extra effort

Acceptance Criteria:

a. User can connect a smartwatch or other bluetooth heart rate device.

b. App can read and retrieve heart rate data during workouts

c. Heart rate data is automatically logged

d. System prevents fake workout entries

Tasks:

a. Implement Bluetooth connection to heart rate sensors

b. Retrieve and process heart rate data in real-time during workouts

c. Auto log heart rate as workouts progress

d. Disable manual logging when auto logging is active to prevent fake entries

10. As someone who wants to track my leg workouts, I want to log squats so that I can see my character get stronger and recover energy faster

Acceptance Criteria:

a. User can select "legs" and then "squats" to log the workout

b. Logging squats increases the leg strength stat

c. Character plays a squat animation when the workout is logged

d. Leg strength stat visually updates on the character (strength/armor appearance)

Tasks:

a. Create a menu to log squats under the "legs" workout category

b. Update leg strength stat and energy recovery rate when squats are logged

c. Animate the character performing squats upon logging

d. Add visual indicators (strength or armor) to reflect increased leg stats

USER STORIES (CONTINUED):

11. As someone who forgets to stay hydrated, I want water reminders during my workout so that I can maintain my performance and health

Acceptance Criteria:

a. User can enable or disable water reminders

b. User can set custom interval timers for reminders

c. Notifications alert the user to drink water during sessions

d. User can mute reminders or pause them with a "Break" option

e. User can confirm water intake or skip the reminder

Tasks:

a. Add a toggle to enable/disable water reminders

b. Implement customizable interval timers for alerts

c. Send notifications with customizable text during workouts

d. Create a "break" button to pause notifications until the user returns

e. Provide options to confirm "intake" or "skip" for each alert

12. As someone who likes personalizing my character, I want to unlock customization
options so that I can change my character's appearance as I progress

Acceptance Criteria:

a. User unlocks character customization at level 10
b. User can apply new outfits or looks once unlocked
c. New styles unlock at specific milestones
d. Custom appearances can be saved to the profile
e. Some items remain locked until earned
f. Character model updates immediately when customization is applied

Tasks:

a. Enable outfit customization when the user reaches level 10
b. Create a menu to apply and change character appearances
c. Unlock additional styles at milestone levels
d. Save and load custom appearances for the character profile
e. Lock certain gear or styles until requirements are met
f. Update the character model dynamically when changes are applied

USER STORIES (CONTINUED):

13. As someone who wants to improve my back strength, I want to log pull-ups so that I can see my character grow stronger with each workout

Acceptance Criteria:

a. User can select "back" and then "pull ups" to log the workout

b. Logging pull ups increases the back strength stat

c. Character performs a pull up animation when the workout is logged

d. Back strength stat updates visually on the character

Tasks:

a. Add a menu option under "back" workouts to log pull-ups

b. Update the back strength stat when pull ups are recorded

c. Play the correct pull up animation on the character upon logging

d. Ensure the character model reflects increased back strength after workouts

14. As someone who likes to track my progress, I want to see a weekly activity summary so
that I can review my workouts, hydration, and boss fights in one place

Acceptance Criteria:

a. User can tap "my week" to view a weekly summary

b. Weekly logs display all workouts, water intake, and boss fights

c. Stats are color coded by day for easy tracking

d. Each day shows accurate and detailed data

e. Daily entries are interactive for more information

Tasks:

a. Create a "my week" section to display weekly activity

b. Generate and update weekly logs automatically

c. Apply color coding to differentiate stats for each day

d. Ensure daily entries display precise workout, water, and boss fight data

e. Make daily entries interactive, allowing users to tap for more details

USER STORIES (CONTINUED):

15. As someone who might miss workouts, I want a rest mode to appear after missing several days so that I can visually see my character resting and feel motivated to return

Acceptance Criteria:

a. Rest mode activates and is visually shown if the user skips 3 or more consecutive days
b. An XP progress bar pauses while rest mode is active
c. App sends a "Come back!" notification to encourage returning
d. Streak resets after 3 missed days, but rest mode remains visible until activity resumes
e. Character displays a clear resting animation or scene during rest mode
f. Rest mode screen provides an option to resume workouts and restart the streak

Tasks:

a. Implement a rest mode screen that appears when the streak is broken
b. Pause XP progression and display a visual indicator (dimmed screen, sleeping character, etc.)
c. Send motivational "Come back!" notifications at set intervals
d. Reset the streak counter after 3 missed days while keeping rest mode active
e. Animate the character in resting state until the user logs a new workout
f. Add a "resume workout" or similar button to exit rest mode and restart progress

Run API Documentation

Create and Play with the Run RPG API

Access Player Profiles, Workout Stats, Leaderboards, and Challenges.

Get Started API Reference

API Documentation

- Getting Started
- Authentication
- Profile
- Input Reviews
- Customer Service
- Free Examples
- Terms of Service

Getting Started

The Run RPG API lets you get game data almost right away. To use it, you need to sign up for a developer account and make a key for your app.

API Keys

API keys help control who can use the API and how much traffic it gets. When you make a key, you have to list the IP addresses that are allowed to use the API. It's best to keep your keys private and make a separate key for each app, so you don't run into problems.

```
This request makes a new API key.

name = The name of your app.

allowed_ips = the IP addresses that are allowed to use this key.
```

Making Requests

To use the API, you need a special access token. You must send this token with every request. The token has rules, like how many times it can be used and which IP addresses are allowed. Because of this, you'll need a web server to get the data and run your app. If you go past the limits, the API will stop working until you're back under them.

```
This request gets a player's info.

Authorization = your access token.

Content Type = tells the system what format the data is in
```

Response Format and Status Codes

All API responses come back in a standard format. They use normal status codes to show if something worked or if there was a problem. If there is an error, the response also includes extra details to explain what went wrong.

```
This is what a successful response looks like

player id = the players ID number

username = the players name

level = the players level in the game

xp = how much experience the player has

stamina = the players energy
```

```
This is what an error response looks like
(401 Unauthorized).

error = Unauthorized

message = why it failed (the access token
is missing or not valid)
```

Available Endpoints

Player Inforamtion

GET

Retrieve player information including stats, level, and progress.

Player Logs

GET

Submit workout data to update player stats and progress.

LeaderBoard

GET

Retrieve global or friend leaderboard rankings.

Profile Customization

GET

Update player profile information and settings.

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